SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I H. Divya Dharshini (192426017 4) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. Dhaadhi

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	1
Method Selection	20%	2	2	2	1	1
Solution Process	25%	2	1	1	2	2
Accuracy of Calculation	20%	2	2	2	2	2
Interpretation of Results	15%	1	1	1	1	1

Total Marks 33 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

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Criteria	Weightage	Q1 8 marks	Q2 8 marks	Q3 8 marks	Q4 8 marks	Q5 8 marks
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Method Selection	20%					
Solution Process	25%					
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Interpretation of Results	15%					

Total Marks **/40** ***Signature of the Faculty***

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Answer (8 Marks)

Given: Network block 192.168.20.0/24 (256 addresses); Lab A = /26, Lab B = /27, Lab C = /28.

Method: Block size = 2^(32 − prefix); usable hosts = 2^(32 − prefix) − 2. Subnets are allocated one after another, starting with the largest, so that they do not overlap.

Step 1: Subnet masks and block sizes

Lab	Prefix	Subnet mask	Host bits	Block size	Usable hosts
A	/26	255.255.255.192	6	64	2^6 − 2 = 62
B	/27	255.255.255.224	5	32	2^5 − 2 = 30
C	/28	255.255.255.240	4	16	2^4 − 2 = 14

Step 2: Allocation

Lab A: starts at 192.168.20.0, covers 64 addresses (.0 to .63)

Lab B: starts at 192.168.20.64, covers 32 addresses (.64 to .95)

Lab C: starts at 192.168.20.96, covers 16 addresses (.96 to .111)

Step 3: Result

Lab	Network address	Valid host range	Broadcast address	Usable hosts
A	192.168.20.0/26	192.168.20.1 – 192.168.20.62	192.168.20.63	62
B	192.168.20.64/27	192.168.20.65 – 192.168.20.94	192.168.20.95	30
C	192.168.20.96/28	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Interpretation: The three subnets are contiguous and do not overlap. Together they use 64 + 32 + 16 = 112 addresses, so 192.168.20.112 – 192.168.20.255 remains free for future labs. The first address of each subnet is the network ID and the last is the broadcast address, so neither can be assigned to a host. Lab A can hold up to 62 devices, Lab B 30 and Lab C 14.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Answer (8 Marks)

Given: System 1 = 192.168.1.130, System 2 = 192.168.1.190, mask = 255.255.255.192.

Step 1: Prefix and block size. 255.255.255.192 = 11111111.11111111.11111111.11000000, which is /26. Host bits = 32 − 26 = 6. Block size = 2^6 = 64. The subnets therefore begin at 0, 64, 128 and 192 in the last octet.

Step 2: Network address by bitwise AND.

System 1: 130 = 10000010; 192 = 11000000; AND = 10000000 = 128 → 192.168.1.128

System 2: 190 = 10111110; 192 = 11000000; AND = 10000000 = 128 → 192.168.1.128

Step 3: Broadcast address. Network + block size − 1 = 128 + 64 − 1 = 191, so the broadcast address is 192.168.1.191 (all host bits set to 1).

Step 4: Result

Parameter	System 1 (192.168.1.130)	System 2 (192.168.1.190)
Subnet range	192.168.1.128 – 192.168.1.191	192.168.1.128 – 192.168.1.191
Network address	192.168.1.128	192.168.1.128
Valid host range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
Broadcast address	192.168.1.191	192.168.1.191
Usable hosts	2^6 − 2 = 62	2^6 − 2 = 62
IP within range?	Yes (129 ≤ 130 ≤ 190)	Yes (129 ≤ 190 ≤ 190)

Interpretation: Both addresses lie in the same subnet, 192.168.1.128/26. System 2 (.190) is the last usable host address, which is valid because .191 is the broadcast address. Since the two systems share one network, they can communicate directly at Layer 2 without a router.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Answer (8 Marks)

Given: 172.16.0.0/16 has 2^16 = 65,536 addresses. Departments are allocated from the largest to the smallest block, sequentially from 172.16.0.0, so that there is no overlap.

Step 1: Block sizes

Department	Prefix	Subnet mask	Block size	Usable hosts
HR	/25	255.255.255.128	128	2^7 − 2 = 126
Sales	/26	255.255.255.192	64	2^6 − 2 = 62
IT	/27	255.255.255.224	32	2^5 − 2 = 30
Admin	/28	255.255.255.240	16	2^4 − 2 = 14

Step 2: Allocation

HR: 172.16.0.0, covers 128 addresses (0 to 127)

Sales: next free = 172.16.0.128, covers 64 addresses (128 to 191)

IT: next free = 172.16.0.192, covers 32 addresses (192 to 223)

Admin: next free = 172.16.0.224, covers 16 addresses (224 to 239)

Step 3: Result

Dept.	Network address	Valid host range	Broadcast	Usable hosts
HR	172.16.0.0/25	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	172.16.0.128/26	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	172.16.0.192/27	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	172.16.0.224/28	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Step 4: Unused address range. Addresses used = 128 + 64 + 32 + 16 = 240, which is 172.16.0.0 to 172.16.0.239. The remaining unused range is 172.16.0.240 – 172.16.255.255, which is 65,536 − 240 = 65,296 addresses.

Interpretation: All four subnets are contiguous and non-overlapping, and they all sit inside 172.16.0.0/24. Almost the entire /16 block remains free for future expansion. Total usable hosts across the departments = 126 + 62 + 30 + 14 = 232.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

Answer (8 Marks)

Given: Network 172.16.10.0/24, subnet mask /27 = 255.255.255.224.

Step 1: Number of subnets. Borrowed bits = 27 − 24 = 3, so subnets = 2^3 = 8.

Step 2: Hosts per subnet. Host bits = 32 − 27 = 5, so block size = 2^5 = 32 and usable hosts = 32 − 2 = 30.

Step 3: List of the 8 subnets

Subnet	Network address	Host range	Broadcast
1	172.16.10.0	172.16.10.1 – .30	172.16.10.31
2	172.16.10.32	172.16.10.33 – .62	172.16.10.63
3	172.16.10.64	172.16.10.65 – .94	172.16.10.95
4	172.16.10.96	172.16.10.97 – .126	172.16.10.127
5	172.16.10.128	172.16.10.129 – .158	172.16.10.159
6	172.16.10.160	172.16.10.161 – .190	172.16.10.191
7	172.16.10.192	172.16.10.193 – .222	172.16.10.223
8	172.16.10.224	172.16.10.225 – .254	172.16.10.255

Step 4: Subnet of 172.16.10.78.

78 = 01001110; 224 = 11100000; AND = 01000000 = 64

Network address = 172.16.10.64

Broadcast address = 64 + 32 − 1 = 95, i.e. 172.16.10.95

Valid host range = 172.16.10.65 – 172.16.10.94

Step 5: Is 172.16.10.95 in the same subnet? Yes. 95 lies between 64 and 95, so it is inside the range 172.16.10.64 – 172.16.10.95 (95 AND 224 = 64, the same network). However, 172.16.10.95 is the broadcast address of this subnet, so it cannot be assigned to a host.

Interpretation: 172.16.10.78 is a valid host in the 3rd subnet (172.16.10.64/27). The address 172.16.10.95 belongs to the same subnet range but is reserved for broadcast; the next address, 172.16.10.96, begins the 4th subnet.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer (8 Marks)

Given: 2001:0db8:0000:0000:0000:ff00:0042:8329 (128 bits, eight 16-bit groups).

Step 1: Compression rules

Rule 1: leading zeros in each group may be dropped. Rule 2: one run of consecutive all-zero groups may be replaced by '::' (used only once per address).

Original: 2001:0db8:0000:0000:0000:ff00:0042:8329

After Rule 1: 2001:db8:0:0:0:ff00:42:8329

After Rule 2: 2001:db8::ff00:42:8329

Shortest form: 2001:db8::ff00:42:8329

Step 2: Expansion

The '::' replaces the missing zero groups. The compressed address has 5 groups written (2001, db8, ff00, 42, 8329), so 8 − 5 = 3 zero groups are restored, and each group is padded with leading zeros to 4 hex digits.

2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329

Expanded form: 2001:0db8:0000:0000:0000:ff00:0042:8329 (identical to the original).

Step 3: Network prefix (/64)

The first 64 bits are the first four groups, 2001:0db8:0000:0000. The remaining 64 bits (0000:ff00:0042:8329) form the interface identifier.

Network prefix: 2001:db8::/64 (full form 2001:0db8:0000:0000::/64).

Step 4: Number of host addresses

Host bits = 128 − 64 = 64, so the number of addresses = 2^64 = 18,446,744,073,709,551,616 (about 1.84 × 10^19).

Interpretation: A /64 is the standard IPv6 subnet size and gives about 18.4 quintillion addresses. IPv6 has no broadcast address, so unlike IPv4 no address is lost to broadcast; only the all-zeros subnet-router anycast address (2001:db8::) is conventionally reserved. The given address is therefore a valid host address in the network 2001:db8::/64.

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